Question 6 - Basic Jenkins CI/CD pipeline

Question 6: Implement a basic Jenkins CI/CD pipeline that pulls source code from the GitHub repository and triggers an automated build.

Repository: https://github.com/ksamirdev/ensp461-devops-lab1

Job configuration

Job text-stats-pipeline uses Pipeline script from SCM, with the public GitHub repository, branch */main and script path Jenkinsfile.

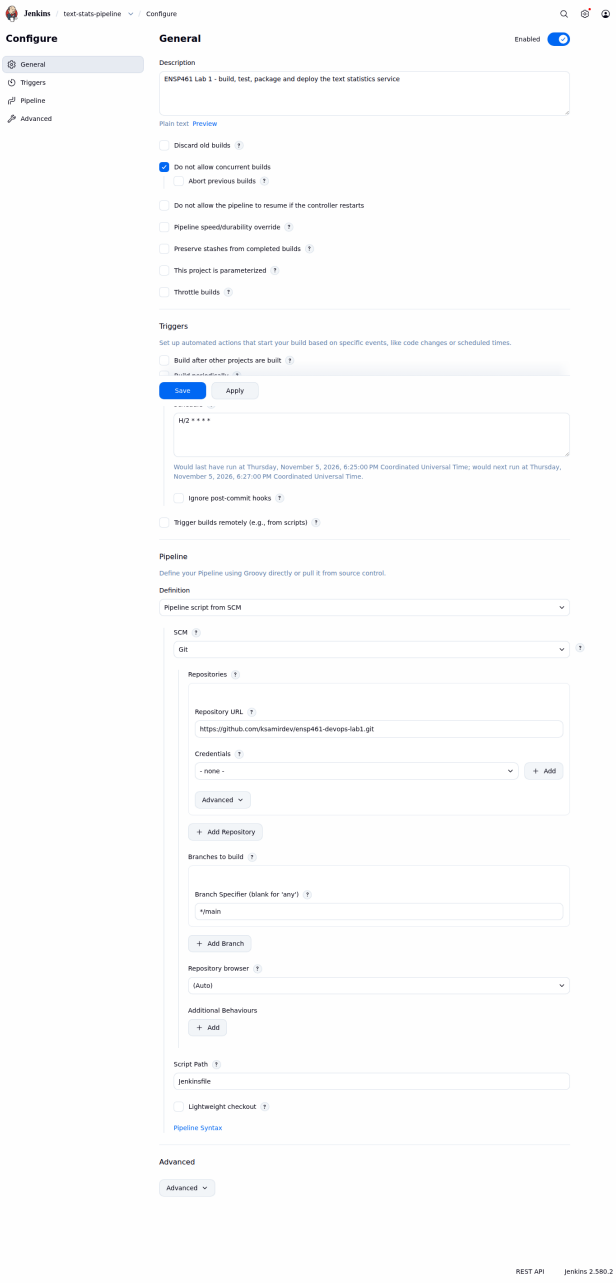


Fig: Pipeline job configuration

Jenkins setup

```
$ sudo mkdir -p /etc/systemd/system/jenkins.service.d

$ printf '[Service]\nEnvironment="JAVA_OPTS=-Djava.awt.headless=true
-Djenkins.install.runSetupWizard=false"\n' | sudo tee /etc/systemd/system/jenkins.service.d/override.conf
[Service]
Environment="JAVA_OPTS=-Djava.awt.headless=true -Djenkins.install.runSetupWizard=false"

$ curl -fsSL -o /tmp/jpm.jar https://github.com/jenkinsci/plugin-installation-manager-
tool/releases/download/2.16.0/jenkins-plugin-manager-2.16.0.jar

$ sudo -u jenkins java -jar /tmp/jpm.jar --war /usr/share/java/jenkins.war -d /var/lib/jenkins/plugins
--plugins workflow-aggregator git junit pipeline-stage-view pipeline-graph-view 2>&1 | tail -3
Done

$ sudo systemctl daemon-reload && sudo systemctl restart jenkins && systemctl is-active jenkins
active

$ sudo install -m 755 ~/ensp461-devops-lab1/deploy/text-stats-deploy /usr/local/bin/text-stats-deploy

$ sudo install -m 644 ~/ensp461-devops-lab1/deploy/text-stats.service /etc/systemd/system/text-stats.service

$ echo 'jenkins ALL=(root) NOPASSWD: /usr/local/bin/text-stats-deploy' | sudo tee /etc/sudoers.d/jenkins-
deploy
jenkins ALL=(root) NOPASSWD: /usr/local/bin/text-stats-deploy

$ sudo chmod 440 /etc/sudoers.d/jenkins-deploy && sudo visudo -c -f /etc/sudoers.d/jenkins-deploy
/etc/sudoers.d/jenkins-deploy: parsed OK

$ sudo mkdir -p /opt/text-stats && sudo systemctl daemon-reload && sudo systemctl enable text-stats
Created symlink /etc/systemd/system/multi-user.target.wants/text-stats.service → /etc/systemd/system/text-
stats.service.
```

Fig: Plugins, service unit and scoped sudo rule (rendered from evidence/logs/q6-jenkins-setup.log)

Build 1

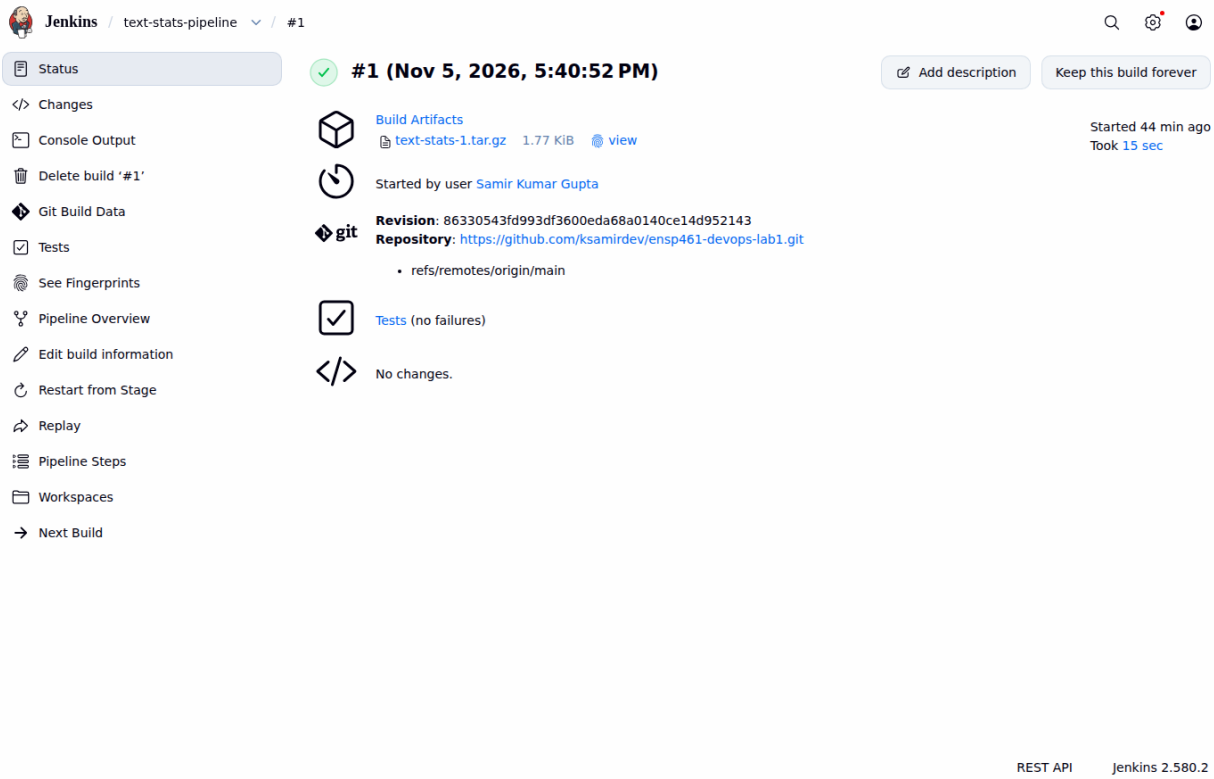


Fig: Build 1 summary

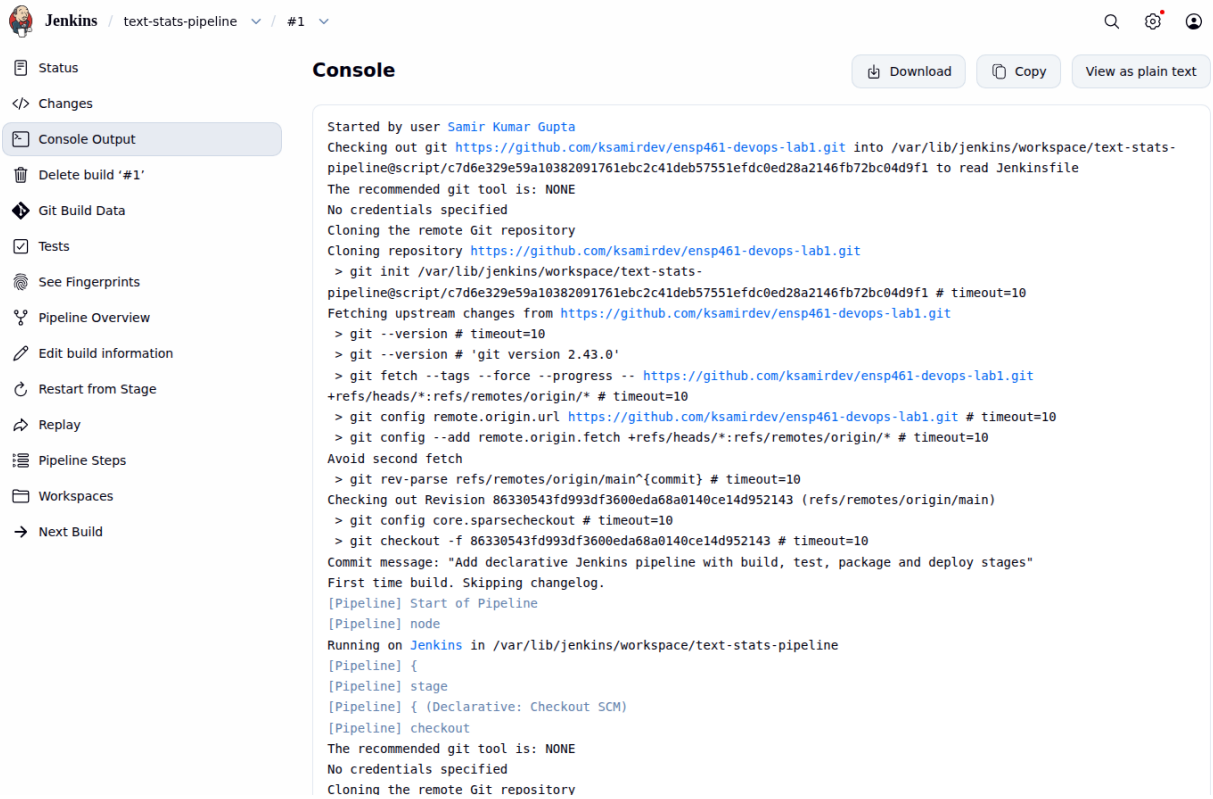


Fig: Build 1 console

```
Started by user Samir Kumar Gupta
Checking out git https://github.com/ksamirdev/ensp461-devops-lab1.git into /var/lib/jenkins/workspace/text-
stats-pipeline@script/c7d6e329e59a10382091761ebc2c41deb57551efdc0ed28a2146fb72bc04d9f1 to read Jenkinsfile
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/ksamirdev/ensp461-devops-lab1.git
Fetching upstream changes from https://github.com/ksamirdev/ensp461-devops-lab1.git
Avoid second fetch
Checking out Revision 86330543fd993df3600eda68a0140ce14d952143 (refs/remotes/origin/main)
Commit message: "Add declarative Jenkins pipeline with build, test, package and deploy stages"
First time build. Skipping changelog.
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/text-stats-pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/ksamirdev/ensp461-devops-lab1.git
Fetching upstream changes from https://github.com/ksamirdev/ensp461-devops-lab1.git
Avoid second fetch
Checking out Revision 86330543fd993df3600eda68a0140ce14d952143 (refs/remotes/origin/main)
Commit message: "Add declarative Jenkins pipeline with build, test, package and deploy stages"
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] sh
+ python3 -m venv .venv
[Pipeline] sh
+ .venv/bin/pip install -q -r requirements.txt
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] sh
+ .venv/bin/python -m pytest -v --junitxml=test-results.xml
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-8.3.4, pluggy-1.7.0 -- /var/lib/jenkins/workspace/text-stats-
pipeline/.venv/bin/python
cachedir: .pytest_cache
rootdir: /var/lib/jenkins/workspace/text-stats-pipeline
collecting ... collected 5 items

tests/test_web.py::test_healthz_reports_healthy PASSED                   [ 20%]
tests/test_web.py::test_text_stats_counts_unique_words_case_insensitively PASSED [ 40%]
tests/test_web.py::test_stats_endpoint_returns_counts PASSED             [ 60%]
tests/test_web.py::test_stats_endpoint_rejects_invalid_text[body0] PASSED [ 80%]
tests/test_web.py::test_stats_endpoint_rejects_invalid_text[body1] PASSED [100%]

- generated xml file: /var/lib/jenkins/workspace/text-stats-pipeline/test-results.xml -
============================== 5 passed in 0.05s ===============================
Post stage
[Pipeline] junit
Recording test results
[Checks API] No suitable checks publisher found.
[Pipeline] stage
[Pipeline] { (Package)
```

Fig: Build 1 console output, Git operations trimmed (rendered from evidence/logs/q6-q7-build1-console.log)

Observations

- Build 1 was started manually and finished with SUCCESS in about 15 seconds.
- Pipeline script from SCM means the pipeline definition is versioned with the code it builds.
- Using a public HTTPS remote means Jenkins needs no stored Git credentials.